Lab 1 Deliverables

Problem Statement #51 **|** *Media, Events & Community — Virtual Technical Conference Platform*

Stakeholders / Actors: Conference Author, Track Chair, Peer Reviewer, Session Attendee

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall anonymize submitted manuscripts (strip author headers and metadata) before assigning them to reviewer pools for double-blind review.	High	Pass: Reviewer sees anonymized paper; Fail: Author identity visible to peer reviewer.	(Given) Core functionality to preserve double-blind review integrity.
FR-002	Functional	The system shall assign each anonymized manuscript to a pool of at least two reviewers who have no declared conflict of interest (e.g., same institution or prior co-authorship) with the submitting author.	High	Pass: No reviewer on a paper's assigned pool appears on that paper's declared conflict-of-interest list; Fail: A conflicted reviewer is assigned.	Conflict-free assignment is essential to reviewer impartiality and process credibility.
FR-003	Functional	The system shall generate a multi-track presentation schedule that assigns every accepted paper to a session slot, room, and track without double-booking any room, track, or speaker.	High	Pass: Automated validation reports zero room, track, or speaker conflicts; Fail: An overlapping booking is detected.	Automates a constraint-satisfaction task that is error-prone to do manually for a multi-track event.
FR-004	Functional	The system shall allow a registered session attendee to submit a text question to the live Q&A queue of the session currently in progress.	Medium	Pass: Submitted question appears in the queue within 3 seconds; Fail: Question fails to appear or errors out.	Enables real-time audience engagement during virtual sessions.
FR-005	Functional	The system shall allow attendees to upvote questions in the live Q&A queue and shall re-sort the queue in descending order of vote count in real time.	Medium	Pass: Queue order updates within 2 seconds of an upvote; Fail: Queue order is stale or unchanged.	Surfaces the most relevant audience questions to speakers and moderators.
NFR-001	Nonfunctional (Performance & Security)	The multi-track schedule generator shall resolve session room capacities and speaker availability constraints in under 10 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	(Given) Ensures a smooth experience during a live, time-boxed conference.
NFR-002	Nonfunctional (Scalability & Reliability)	The system shall support at least 500 concurrent Q&A upvote actions per session and shall maintain 99.5% platform uptime during scheduled conference hours.	High	Pass: Load test of 500 concurrent upvotes completes within 2 seconds with zero dropped requests, and uptime meets 99.5%; Fail: Requests dropped or uptime below target.	The platform is the sole venue for a live, time-critical event; downtime or lag directly disrupts the experience.

2. Use-Case Flow Specification

Use Case Name: UC-03 - Assign Reviewers to Paper

Primary Actor: Track Chair
Secondary Actors: Peer Reviewer (receives assignment notification), Conference Author (paper owner, indirectly affected)
System: Virtual Technical Conference Platform
Includes: UC-02 Anonymize Manuscript (always executed as part of UC-01 Submit Manuscript, before this use case begins)

1. Preconditions

- The manuscript has been submitted and successfully anonymized (author headers and metadata stripped) by the system (UC-02).
- The Track Chair is authenticated and has an active session with track-management privileges.
- A pool of registered peer reviewers with declared subject-area expertise and conflict-of-interest declarations exists for the paper's track.

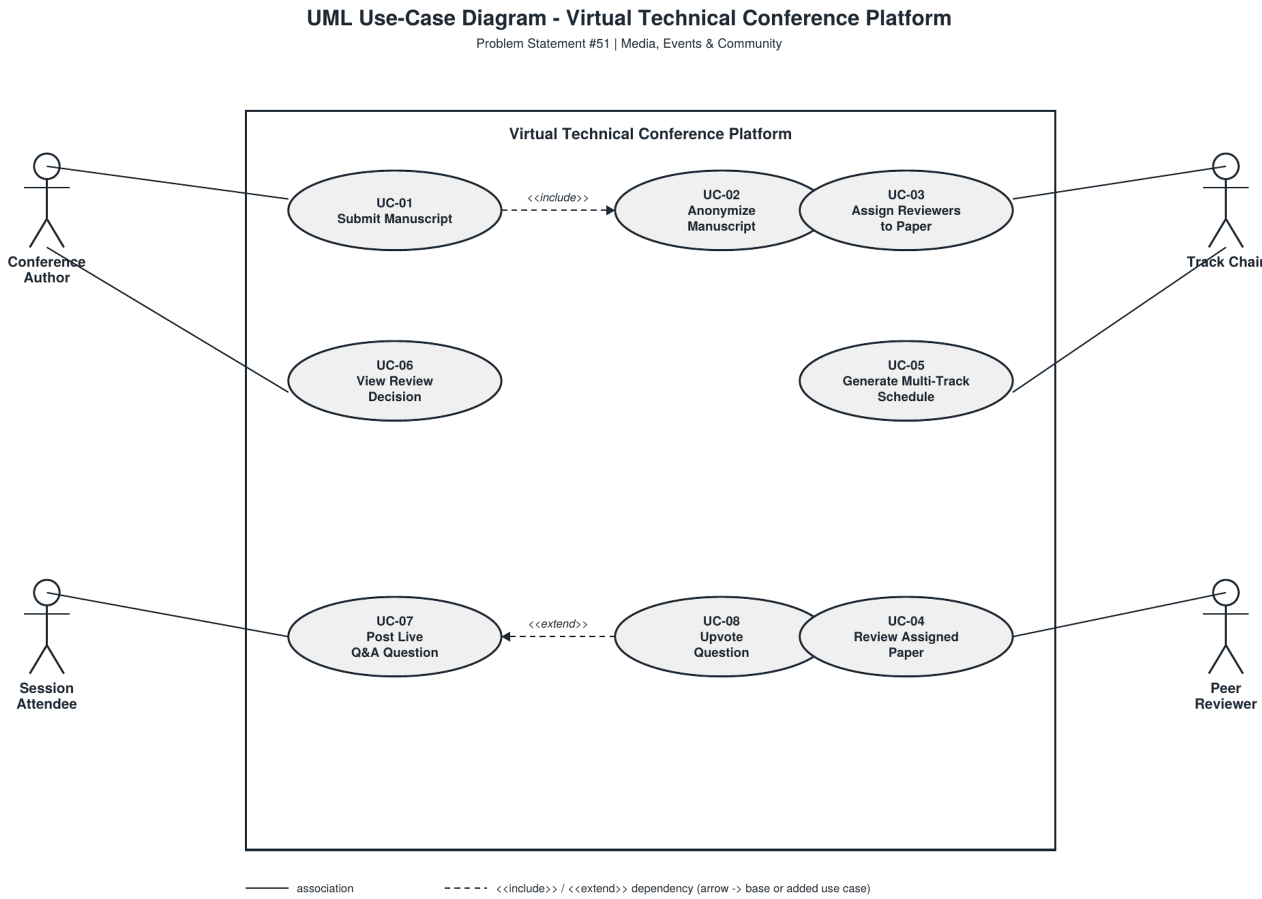
2. Postconditions

- Success: The paper is linked to at least two reviewers with no declared conflict of interest, its status changes to “Under Review,” and each assigned reviewer is notified.
- Failure: The paper remains in “Pending Assignment” status and the Track Chair is notified that manual intervention is required.

3. Main Success Scenario

1. The Track Chair selects “Assign Reviewers” for a paper from the Pending Assignment queue.
2. The System displays the anonymized manuscript's track, topic tags, and abstract (no author-identifying data).
3. The System computes a candidate reviewer list by matching topic tags to reviewer expertise and filtering out any reviewer on the paper's conflict-of-interest list.
4. The System presents the ranked candidate list to the Track Chair, showing each reviewer's current review load.
5. The Track Chair selects at least two reviewers from the candidate list and confirms the assignment.
6. The System validates that none of the selected reviewers appear on the conflict-of-interest list.
7. The System records the assignment, updates the paper status to “Under Review,” and starts the review-deadline timer.
8. The System sends a notification (email/in-app) to each assigned reviewer with a link to the anonymized manuscript.
9. The use case ends successfully.

3. UML Use-Case Diagram



4. Alternate Flow

6a. Conflict of Interest Detected

1. During assignment confirmation (Step 6), the System detects that a reviewer selected by the Track Chair appears on the paper's conflict-of-interest list.
2. The System rejects the assignment for that reviewer and displays a warning identifying the conflicting reviewer.
3. The System returns the Track Chair to the candidate list (Step 4) with the conflicting reviewer removed from the selectable pool.
4. The Track Chair selects a replacement reviewer and resubmits; flow resumes at Step 6.
5. If the Track Chair cannot find enough eligible reviewers after 3 attempts, the paper is flagged “Needs Manual Assignment” and the use case ends in the failure postcondition.